

Eigenvalues and Eigenvectors

For a square matrix A , when Ax is parallel to x i.e., $Ax = \lambda x$. We can say λ is the eigenvalue and x is the eigenvector of matrix A . If A is singular, then x is in nullspace and $\lambda=0$.

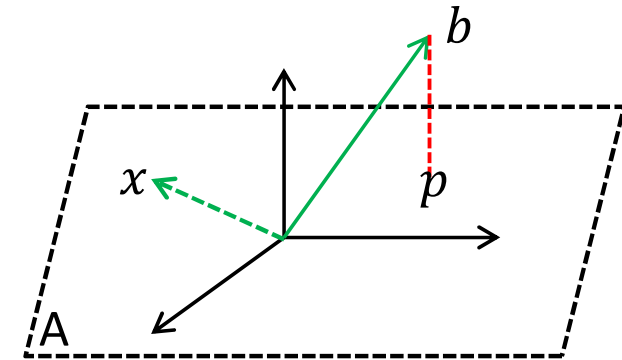
What are the eigenvalues and eigenvectors for a projection matrix P ?

Let the projection matrix P projects all vectors into plane A .

From the picture we can say that b is not an eigenvector ($Pb \neq b$).

Any vector x in the plane A , $Px = x$ is valid.

So, any vectors in the plane A is eigenvector and their eigenvalue is 1.



What will be the independent eigenvectors of the projection matrix P ?

Corresponding to this figure we will have 3 independent eigenvectors. Two of them are in the plane A and the rest one is in nullspace of P , i.e., any vector which is perpendicular to

Plane A and its eigenvalue is 0 (as $P \cdot x = 0 \cdot x$).

Example: Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

For $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $Ax = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = x$, So eigenvalue is 1

For $x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $Ax = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -x$, So eigenvalue is -1

Facts:

sum of λ 's = sum of $a_{11}, a_{22}, \dots, a_{nn}$.
multiplication of λ 's = $\det(A)$.

Eigenvalues and Eigenvectors

How to compute eigenvalues and eigenvectors?

If λ and x are the eigenvalue and eigenvector of A then $Ax = \lambda x$.

$$Ax - \lambda x = 0$$

$$(A - \lambda I)x = 0$$

This tells us that subtracting λ from the diagonal element of A makes the matrix $(A - \lambda I)$ singular. Thus, $\det(A - \lambda I) = 0$.

The equation $\det(A - \lambda I) = 0$ is called the characteristic equation. Solving this equation first we will get eigenvalues then for each eigenvalue the corresponding eigenvector is computed.

Example: Find the λ 's and x 's for $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

$$\det(A - \lambda I) = 0 \quad \longrightarrow \quad \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{vmatrix} = 0$$
$$(3 - \lambda)^2 - 1 = 0$$
$$\lambda^2 - 6\lambda + 8 = 0$$
$$(\lambda - 4)(\lambda - 2) = 0$$

So, $\lambda_1 = 4$ and $\lambda_2 = 2$.

For $\lambda_1 = 4$, the matrix $A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ is singular.

So, $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0$, when x is in nullspace.

So, $x_1 = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} = U \right)$

For $\lambda_2 = 2$, the matrix $A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is singular.

So, $x_2 = c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = U \right)$

The trace of A is $\lambda_1 + \lambda_2 = 6$,
 $\det(A) = \lambda_1 \cdot \lambda_2 = 8$

So, for $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, $\lambda_1 = 4$, $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
and $\lambda_2 = 2$, $x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Eigenvalues and Eigenvectors

If we add a scalar constant c to the diagonal elements of a matrix, its eigenvalues will increase by c but its eigenvectors remain same.

Proof: Let $Ax = \lambda x$,

$$\begin{aligned}(A + cI)x &= Ax + cx \\ &= \lambda x + cx \\ &= (\lambda + c)x\end{aligned}$$

Example: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ have same eigenvectors but the eigenvalues of $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are $\lambda_1 = 1$ and $\lambda_2 = -1$.

Whereas, the eigenvalues of $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ are $\lambda_1 = 4$ and $\lambda_2 = 2$.

Example: Find the λ 's and x 's for $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

$$\det(A - \lambda I) = 0 \quad \Rightarrow \quad \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda_1 = i, \lambda_2 = -i.$$

For $\lambda_1 = i$, the matrix $A - iI = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}$ is singular.

$$\text{So, } x_1 = c \begin{bmatrix} i \\ 1 \end{bmatrix}, \quad \left(\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} = \begin{bmatrix} -i & -1 \\ 0 & 0 \end{bmatrix} = U \right)$$

For $\lambda_2 = -i$, the matrix $A + iI = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix}$ is singular.

$$\text{So, } x_2 = c \begin{bmatrix} -i \\ 1 \end{bmatrix}, \quad \left(\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} = \begin{bmatrix} i & -1 \\ 0 & 0 \end{bmatrix} = U \right)$$

The trace of A is $i - i = 0$,
 $\det(A) = -i \cdot i = 1$

So, for $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\lambda_1 = i, x_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$
 and $\lambda_2 = -i, x_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$.

Eigenvalues and Eigenvectors

Example: Find the λ 's and x 's for $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.

$$\det(A - \lambda I) = 0 \quad \longrightarrow \quad \begin{vmatrix} 3 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix} = 0$$
$$(3 - \lambda)^2 = 0$$
$$\lambda_1 = 3, \lambda_2 = 3.$$

The trace of A is $3 + 3 = 6$,
 $\det(A) = 3 \cdot 3 = 9$

For $\lambda_1 = 3$, the matrix $A - 3I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is singular.

$$\text{So, } x_1 = c \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

For $\lambda_2 = 3$, $x_2 = c \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. So, no 2nd independent eigenvector found. This matrix is called degenerate matrix.

Diagonalization and power of A

Suppose we have n independent eigenvectors of A . Putting them in the columns a matrix S is formed.

$$AS = A \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ x_1 & x_2 & \dots & x_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ \lambda_1 x_1 & \lambda_2 x_2 & \dots & \lambda_n x_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ x_1 & x_2 & \dots & x_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} = S\Lambda$$

So, $AS = S\Lambda$
 $\Rightarrow S^{-1}AS = \Lambda$
 $\Rightarrow A = S\Lambda S^{-1}$

$\uparrow$

diagonal eigenvalue matrix

Theorem: Eigenvalue of A^k is k^{th} power of λ (i.e., λ^k). The eigenvectors of A and A^k are same.

Proof:

If $Ax = \lambda x$, then $A^2x = A\lambda x = \lambda Ax = \lambda^2 x$
 $A^k x = \lambda^k x$

another way: $A^2 = S\Lambda S^{-1} S\Lambda S^{-1} = S\Lambda^2 S^{-1}$
 $A^k = S\Lambda^k S^{-1}$

Theorem: If all $|\lambda_i| < 1$, then $A^k \rightarrow 0$, when $k \rightarrow \infty$.

Are all matrices diagonalizable?

A matrix is diagonalizable if it has n independent eigenvectors. A matrix sure to have n independent eigenvectors and be diagonalizable if all the λ 's are different (i.e., no repeated λ 's).

Repeated eigenvalues may or may not have n independent eigenvectors!

For identity matrices I , its all λ 's are 1 but it has n independent eigenvectors.

$$(S^{-1}AS = S^{-1}IS = S^{-1}S = I).$$

For degenerate matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ have eigenvalues 3 and 3. Its only eigenvector is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. 2nd independent eigenvector is not found.

Difference equation and power of A

Difference equations $u_{k+1} = Au_k$ move forward in a finite number of finite steps. Start with a given vector u_0 , we get:

$$u_1 = Au_0; \quad u_2 = A^2u_0; \quad \dots \quad ; \quad u_{k+1} = A^{k+1}u_0.$$

So, the solution to a difference equation $u_{k+1} = Au_k$ is $u_k = A^k u_0$.

To really solve this, we rewrite: $u_0 = c_1x_1 + c_2x_2 + \dots + c_nx_n = SC$

$$\text{Now,} \quad u_1 = Au_0 = c_1Ax_1 + c_2Ax_2 + \dots + c_nAx_n = ASC$$

$$= c_1\lambda_1x_1 + c_2\lambda_2x_2 + \dots + c_n\lambda_nx_n = S\Lambda C$$

$$u_2 = A^2u_0 = c_1\lambda_1Ax_1 + c_2\lambda_2Ax_2 + \dots + c_n\lambda_nAx_n = AS\Lambda C$$

$$= c_1\lambda_1^2x_1 + c_2\lambda_2^2x_2 + \dots + c_n\lambda_n^2x_n = S\Lambda^2C$$

$$u_k = A^k u_0 = c_1\lambda_1^kx_1 + c_2\lambda_2^kx_2 + \dots + c_n\lambda_n^kx_n = S\Lambda^kC$$

Example: Find the 100th Fibonacci number.

Fibonacci numbers: 0, 1, 1, 2, 3, 5, 8, 13,

Fibonacci equation: $F_{k+2} = F_{k+1} + F_k$.

The goal is to solve the difference equation $F_{k+2} = F_{k+1} + F_k$. This can be reduced to a one-step equation

$u_{k+1} = Au_k$. Every step multiplies $u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$ by a matrix A:

$$F_{k+2} = F_{k+1} + F_k \quad \longrightarrow \quad u_{k+1} = \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = Au_k \text{ where } u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$$F_{k+1} = F_{k+1}$$

So, the 100th Fibonacci number will obtain by computing $u_{100} = S\Lambda^{100}C$.

Difference equation and power of A

Computing the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \text{ and } \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Now $\begin{bmatrix} 1 - \lambda_1 & 1 \\ 1 & -\lambda_1 \end{bmatrix} [x_1] = 0$, when $x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 - \lambda_2 & 1 \\ 1 & -\lambda_2 \end{bmatrix} [x_2] = 0$, when $x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$.

As we know $c_1 x_1 + c_2 x_2 = u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$c_1 \lambda_1 + c_2 \lambda_2 = 1 \text{ and } c_1 + c_2 = 0$$

$$c_1 = \frac{1}{\sqrt{5}} \text{ and } c_2 = -\frac{1}{\sqrt{5}}.$$

So, the 100th Fibonacci number is $u_{100} = A^{100} u_0 = S \Lambda^{100} C$

$$= \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^{100} & 0 \\ 0 & \lambda_2^{100} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}$$

Markov Matrix

A matrix is called Markov matrix if its

- i) All entries ≥ 0 .
- ii) All columns add to 1.
- iii) Power is also a Markov matrix.

$$A = \begin{bmatrix} .1 & .01 & .3 \\ .2 & .99 & .3 \\ .7 & 0 & .4 \end{bmatrix} \text{ is a Markov matrix}$$

Key Points:

- 1. $\lambda=1$, is an eigenvalue.
- 2. All other λ 's are between -1 to 1, i.e., $|\lambda| < 1$.

The solution of difference equation $u_{k+1} = Au_k$ for Markov matrix A is:

$$u_k = A^k u_0 = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \dots + c_n \lambda_n^k x_n$$

Let $\lambda_1 = 1$ and rest $|\lambda_i| < 1$, then for $k \rightarrow \infty$, $u_k = c_1 x_1$. Thus the solution reach in steady state.

One eigenvalue of Markov matrix is 1:

$|A - I| = \begin{bmatrix} -.9 & .01 & .3 \\ .2 & -.01 & .3 \\ .7 & 0 & -.6 \end{bmatrix}$ Since all columns of $A - I$ add to zero, the matrix $A - I$ is singular. So the vector present in the nullspace of $A - I$ is the eigenvector of A.

Application: $u_{k+1} = Au_k$, A is a Markov matrix.

$$\begin{aligned} \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=k+1} &= \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=k}, \quad \text{Let } \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 1000 \end{bmatrix} \\ \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=1} &= A \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=0} = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} 0 \\ 1000 \end{bmatrix} = \begin{bmatrix} 200 \\ 800 \end{bmatrix}, \quad \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=2} = A \begin{bmatrix} u_{cal} \\ u_{mad} \end{bmatrix}_{t=1} = \\ &\quad \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} 0 \\ 1000 \end{bmatrix} = \begin{bmatrix} 340 \\ 660 \end{bmatrix}, \dots \end{aligned}$$

Markov Matrix

Compute the eigenvalues and eigenvectors:

Since the matrix $\begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix}$ is a Markov matrix, its one eigenvalue $\lambda_1 = 1$.

For $\lambda_1 = 1$, $\begin{bmatrix} -.1 & .2 \\ .1 & -.2 \end{bmatrix} \begin{bmatrix} \vdots \\ x_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. So the corresponding eigenvector $x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

For other eigenvalue $\lambda_2 = 1.7 - 1 = .7$, $\begin{bmatrix} .2 & .2 \\ .1 & .1 \end{bmatrix} \begin{bmatrix} \vdots \\ x_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. So the corresponding eigenvector $x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Since, $SC = u_0$

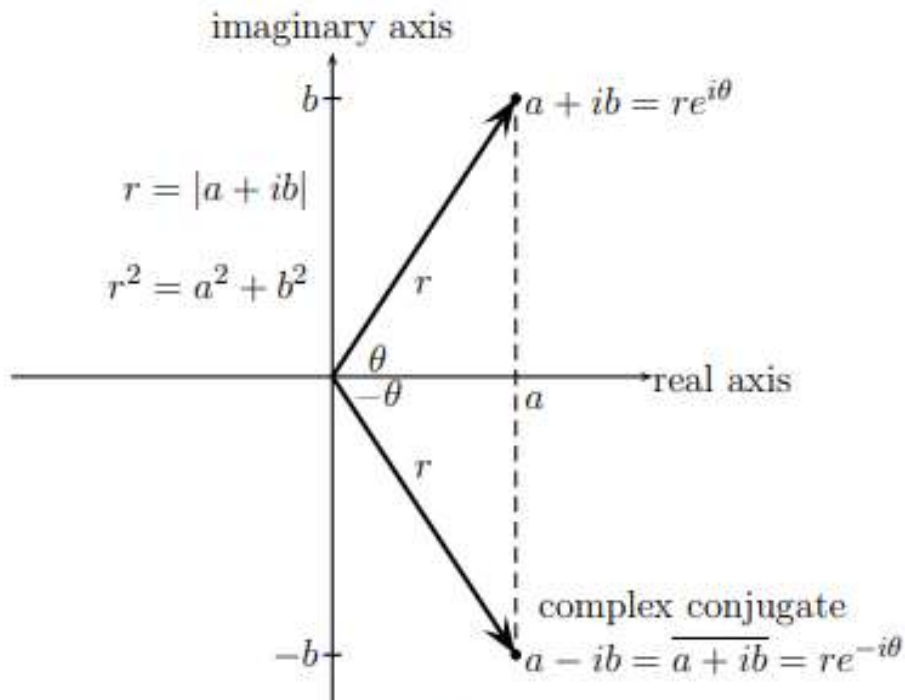
$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1000 \end{bmatrix}$$

$$c_1 = \frac{1000}{3} \text{ and } c_2 = \frac{2000}{3}.$$

Now,
$$u_k = A^k u_0 = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2$$
$$= \frac{1000}{3} (1)^k x_1 + \frac{2000}{3} (.7)^k x_2$$

For steady state,
$$u_{k=\infty} = \frac{1000}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2000}{3} \\ \frac{1000}{3} \end{bmatrix}.$$

Complex matrices



The complex plane, with $a + ib = re^{i\theta}$ and its conjugate $a - ib = re^{-i\theta}$.

Length of complex vector:

Let $z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ in C^n space. Now $\|z\|^2 = z^T z$ is not good,

But $\|z\|^2 = \bar{z}^T z$ is good.

Example: Let $z = \begin{bmatrix} 2 \\ i \end{bmatrix}$,

$\|z\|^2 = \begin{bmatrix} 2 & i \end{bmatrix} \begin{bmatrix} 2 \\ i \end{bmatrix} = 2^2 + i^2 = 3$ is not good.

$\|z\|^2 = \begin{bmatrix} 2 & -i \end{bmatrix} \begin{bmatrix} 2 \\ i \end{bmatrix} = 2^2 - i^2 = 5$ is good.

So, length of a complex vector $z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ is computed as $\|z\|^2 = \bar{z}^T z = [\bar{z}_1 \ \bar{z}_2 \ \dots \ \bar{z}_n] \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = \bar{z}_1 z_1 + \bar{z}_2 z_2 + \dots + \bar{z}_n z_n$. It is also defined as $\|z\|^2 = z^H z$, where $z^H = \bar{z}^T$. H stand for Hermitian.

Example: Compute the length of $y = \begin{bmatrix} 2 + i \\ 2 - 4i \end{bmatrix}$.

$$\|y\|^2 = y^H y = \bar{y}^T y = [2 - i \quad 2 + 4i] \begin{bmatrix} 2 + i \\ 2 - 4i \end{bmatrix} = 5 + 20 = 25$$

Complex matrices

Inner product: For complex vectors X and Y , the inner product is computed as $\bar{Y}^T X = Y^H X$.

Symmetric matrix: If A is a complex matrix and it will be symmetric if $\bar{A}^T = A$. Note that $A^T = A$ is not good if A is complex.

Example: Let $A = \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$, Now $A^T = \begin{bmatrix} 2 & 3-i \\ 3+i & 5 \end{bmatrix} \neq A$ But $A^H = \bar{A}^T = \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix} = A$. Here A is called Hermitian matrix.

Perpendicular complex vector(unitary matrix): Let $q_1, q_2, \dots, q_n$ are orthonormal vectors then

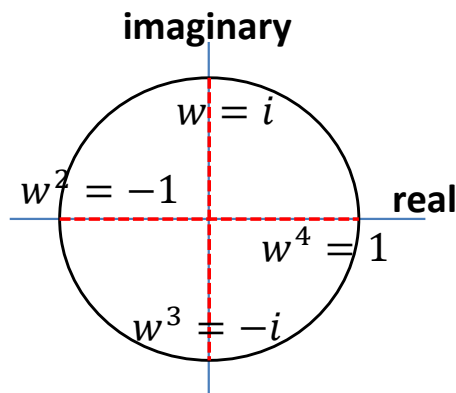
$$\bar{q}_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$\text{For } Q = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ q_1 & q_2 & \dots & q_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix}, \quad \bar{Q}^T Q = Q^H Q = I$$

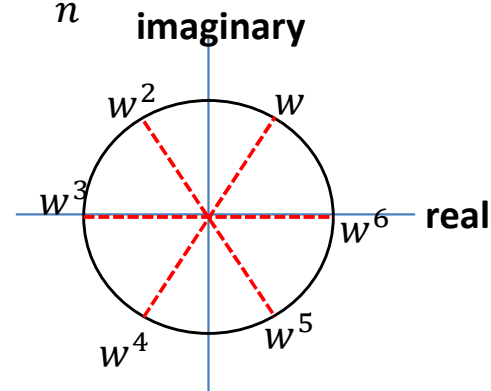
Fourier Matrix: columns are orthogonal

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & \dots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \dots & w^{(n-1)^2} \end{bmatrix}$$

So, $(F_n)_{ij} = w^{ij}$, $i, j = 0, 1, \dots, n-1$. Where w is a spacial number, such that $w^n = 1$. So, $w = e^{i\frac{2\pi}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$.



$$\begin{aligned} \text{For, } n &= 4, \\ w &= e^{i\frac{2\pi}{n}} \\ &= i \end{aligned}$$



$$\begin{aligned} \text{For, } n &= 6, \\ w &= e^{i\frac{2\pi}{n}} \\ &= e^{i\frac{\pi}{3}} \end{aligned}$$

Complex matrices

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w^6 \\ 1 & w^3 & w^6 & w^9 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

$$\text{So, } F_4^H F_4 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$$

Symmetric matrices

Symmetric matrices: $A = A^T$ or $A = A^H$

Properties:

- i) The eigenvalues are **REAL**.
- ii) The eigenvectors are (can be chosen) **PERPENDICULAR**.

Eigenvalues of a symmetric matrix are real:

If A is a symmetric matrix and λ and x are its eigenvalues and eigenvectors, then $Ax = \lambda x$. Now if $\bar{\lambda}$ and $\bar{x}$ represents the conjugate of λ , and x , respectively, then $A\bar{x} = \bar{\lambda}\bar{x}$ is true.

Now multiplying $\bar{x}^T$ both side of $Ax = \lambda x$, we get $\bar{x}^T Ax = \lambda \bar{x}^T x$.

$$\begin{aligned}
 \text{So, } A\bar{x} = \bar{\lambda}\bar{x} &\quad \Rightarrow (A\bar{x})^T = (\bar{\lambda}\bar{x})^T \\
 &\quad \Rightarrow \bar{x}^T A^T = \bar{x}^T \bar{\lambda}^T \\
 &\quad \Rightarrow \bar{x}^T A = \bar{x}^T \bar{\lambda} \quad (\text{If } A \text{ is real, then } \bar{A} = A \text{ and if } A \text{ is complex, then } \bar{A}^T = A.) \\
 &\quad \Rightarrow \bar{x}^T Ax = \bar{\lambda} \bar{x}^T x \\
 &\quad \Rightarrow \lambda \bar{x}^T x = \bar{\lambda} \bar{x}^T x \quad (\bar{x}^T x = \|x\|^2) \\
 &\quad \Rightarrow \bar{\lambda} = \lambda. \text{ This shows that } \lambda \text{ is real.}
 \end{aligned}$$

Square matrix A can be represented as $A = S \Lambda S^{-1}$.

If A is a symmetric then $A = S \Lambda S^{-1} = Q \Lambda Q^{-1} = Q \Lambda Q^T$ (known as spectrum theorem)

$$A = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ q_1 & q_2 & \dots & q_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \begin{bmatrix} \dots & q_1^T & \dots \\ \dots & q_2^T & \dots \\ \dots & \dots & \dots \\ \dots & q_n^T & \dots \end{bmatrix} = \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \dots + \lambda_n q_n q_n^T$$

All symmetric matrices are combinations of one-dimensional projections—which are symmetric matrices of rank 1.

Symmetric matrices

Example: We already computed the eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ are:

$\lambda_1 = 3, x_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \lambda_2 = 1, x_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Now according to the spectrum theorem, A can be represented as:

$$\begin{aligned} A &= \lambda_1 x_1 x_1^T + \lambda_2 x_2 x_2^T = 3 \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} + 1 \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= 3 \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \end{aligned}$$

The matrices on the right-hand side are $x_1 x_1^T$ and $x_2 x_2^T$ are projections onto the line through x_1 and the line through x_2 .

Facts:

For symmetric matrices :

i) signs of pivots same as signs of eigenvalues, i.e., number of positive pivots = number of positive eigenvalues and number of negative pivots = number of negative eigenvalues.

ii) Product of pivots = product of eigenvalues = determinant.